# SpectralDINO: Dual Mixture-of-Subspaces Low-Rank Adaptation for Cross-Domain Hyperspectral Image Few-Shot Classification

Baocheng Chen, Geng Zhang, Tieqiao Chen, Meng Wang, Jia Liu, Yihao Wang, Renhao Zhang, Siyuan Li, and Bingliang Hu

***Abstract*—Recently, few-shot learning (FSL)-based methods have achieved impressive results in cross-domain hyperspectral classification. However, existing approaches often ignore differences in spectral information caused by varying spectral range across different datasets and encounter limitations due to the constrained parameter size of the models. Furthermore, the substantial differences between RGB and hyperspectral images (HSIs) present significant challenges in applying foundation models (e.g., SAM and DINOv2) to the hyperspectral domain. This article proposes a novel framework for cross-domain few-shot hyperspectral classification that leverages parameter-efficient fine-tuning (PEFT), which we apply to DINOv2 to construct SpectralDINO. Different spectral ranges reflect different physical properties of the target. To enhance the consistency of the spectral features extracted by the model from different domains, we introduce a source-domain spectral alignment (SDSA) strategy to align the spectral ranges and bands of the source-domain data to the target domain. SpectralDINO employs the visual foundation model to enhance its ability to generalize cross-domain knowledge. Additionally, we propose a dual mixture-of-subspaces low-rank adaptation (Dual-MoS LoRA) method to address the structural limitation of the LoRA methods in distinguishing domain-specific features from multidomain inputs. Only 1.14% of the 21.37 M parameters need to be trained to perform fine-tuning. Extensive experimental results on three public datasets demonstrate the superiority of SpectralDINO.**

***Index Terms*—Cross-domain, few-shot learning (FSL), foundation models, hyperspectral image classification (HSIC), parameter-efficient fine-tuning (PEFT).**

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Baocheng Chen and Renhao Zhang are with the Key Laboratory of Spectral Imaging Technology, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an 710119, China, and also with the University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: chenbaocheng2022@opt.ac.cn; zhangrenhao2022@opt.ac.cn).

Geng Zhang, Tieqiao Chen, Yihao Wang, Siyuan Li, and Bingliang Hu are with the Key Laboratory of Spectral Imaging Technology, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an 710119, China (e-mail: gzhang@opt.ac.cn; chentieqiao@opt.ac.cn; wangyihao@opt.ac.cn; lsy@opt.ac.cn; hbl@opt.ac.cn).

Meng Wang is with the Centre for Innovation and Precision Eye Health, the Department of Ophthalmology, and the Yong Loo Lin School of Medicine, National University of Singapore, Singapore 119228 (e-mail: wangmeng9218@126.com).

Jia Liu is with the Key Laboratory of Spectral Imaging Technology, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an 710119, China, also with the University of Chinese Academy of Sciences, Beijing 100049, China, and also with the State Key Laboratory of Tropical Oceanography, South China Sea Institute of Oceanology, Chinese Academy of Sciences, Guangzhou 510301, China (e-mail: liujia1@opt.ac.cn).

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## I. Introduction

HYPERSPECTRAL images (HSIs) are three-dimensional data cubes comprising tens or even hundreds of channels, containing rich spatial and spectral information. Since spectra can reflect the physical properties of features, HSI analysis has been widely used in various fields, such as mineral exploration [1], precision agriculture [2], [3], and environmental monitoring [4].

HSI classification (HSIC) is a crucial task in HSI analysis, aiming to assign labels to each pixel in the HSI. Existing HSIC methods typically train and test models on a single-scene dataset and rely on a large amount of labeled data. However, in real remote sensing applications, HSIC algorithms often encounter newly acquired HSIs in unfamiliar scenes. Moreover, due to the costly and challenging nature of labeling hyperspectral data, newly acquired HSIs often lack sufficient labeled samples. This constraint limits the effectiveness of deep learning-based HSIC methods in practical applications [5].

Recent approaches have focused on achieving high-accuracy HSIC with a limited number of training samples. These methods typically use samples in a well-annotated dataset (source domain) and the few labeled samples in the dataset to be classified (target domain) to train the model and then perform classification on the target domain. This task is called cross-domain few-shot HSIC [6], or cross-domain HSIC for short. Since a few labeled samples from the target domain are used for model training, cross-domain HSIC is essentially a supervised cross-domain classification task. The significant domain shift between source- and target-domain data presents

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a major challenge for cross-domain HSIC. Atmospheric conditions and sensor differences often result in the source-domain data used for training being substantially different from the target-domain data [7], [8]. These disparities manifest in various aspects, such as image acquisition location, land-cover classes, spectral range, number of bands, and spatial resolution. Although existing cross-domain HSIC methods strive to reduce the domain shift between the source and target domains through domain adaptation and domain generalization, most of these methods overlook the heterogeneity of spectral information across different datasets. Each band of HSIs exhibits specific reflection patterns for distinct features [9]. If the source domain provides excessive information about bands that do not exist in the target domain, the network may overemphasize spectral features that cannot be extracted in the target domain, leading to diminished classification results. Conversely, if the source domain offers only partial band information in the spectral range of the target domain, the network may fail to fully capture important spectral features in the target domain.

Another major challenge for cross-domain HSIC is the lack of sufficient training samples in the target domain. Recently, few-shot learning (FSL) has been introduced to address this problem. FSL is a meta-learning method that employs an episode-learning strategy for meta-training. Unlike traditional deep learning, the goal of FSL is to enable the model to learn to distinguish between samples, rather than directly learning what the samples are [10]. Among the FSL frameworks used to solve cross-domain few-shot classification problems, prototype-based FSLs have attracted significant attention due to their simplicity and effectiveness [11], [12], [13], [14]. Specifically, prototype-based FSL predicts the class probability of a sample by measuring the Euclidean distance between the sample features and the prototypes of each class to assess their similarity. Liu et al. [15] applied a prototype network to cross-domain few-shot HSIC using a 3-D CNN as a feature extraction network and learned the metric space from four source-domain datasets. Addressing the limitations of Euclidean distance assumptions, Xi et al. [12] introduced the Mahalanobis distance and proposed an FSL based on the class covariance metric. Liu et al. [13] combined supervised contrastive learning and FSL for the first time and proposed a triple constraints-based prototype learning module to obtain more accurate prototypes with insufficient samples. Due to the scarcity of training samples, models with large parameter sizes are prone to overfitting during training. Therefore, some approaches focus on designing lightweight feature extraction networks to reduce the model size and better accommodate the limited training data. Xue et al. [16] proposed S3Net, which employs 1-D convolution and 2-D convolution instead of 3-D convolution, and global average pooling (GAP) instead of the fully connected layer, significantly reducing the parameter size. However, reducing the model size also constrains the learning capability of the model, making it difficult for the model to learn cross-domain universal features from the source domain.

Recently, with the proposal of visual foundation models such as MAE [17], CLIP [18], SAM [19], and DINOv2 [20], there has been significant interest in adapting these models to downstream tasks [21], [22]. However, as the size of these models continues to increase, the cost of fine-tuning has become prohibitively expensive. To address this issue, researchers have proposed parameter-efficient fine-tuning (PEFT) [23]. Low-rank adaptation (LoRA) [24] is a representative approach of the PEFT technique to adapt the model to a new task by freezing the pretrained base model and using a pair of small fine-tuned components consisting of low-rank projection matrices, the structure of which is shown in Fig. 1(a). Wu et al. [25] introduced a subspace-inspired mixture-of-subspaces LoRA (MoSLoRA) method, which incorporates learnable mixing matrices to enhance subspace fusion effectiveness, as shown in Fig. 1(b). To enhance the model's ability to learn cross-domain universal knowledge, we explore the application of PEFT to the foundation model for adaptation to the cross-domain HSIC task. Implementing PEFT can help alleviate overfitting caused by insufficient samples by keeping the training parameter size small. Simultaneously, the foundation model retains generalization knowledge from the natural image domain and provides stronger learning capabilities.

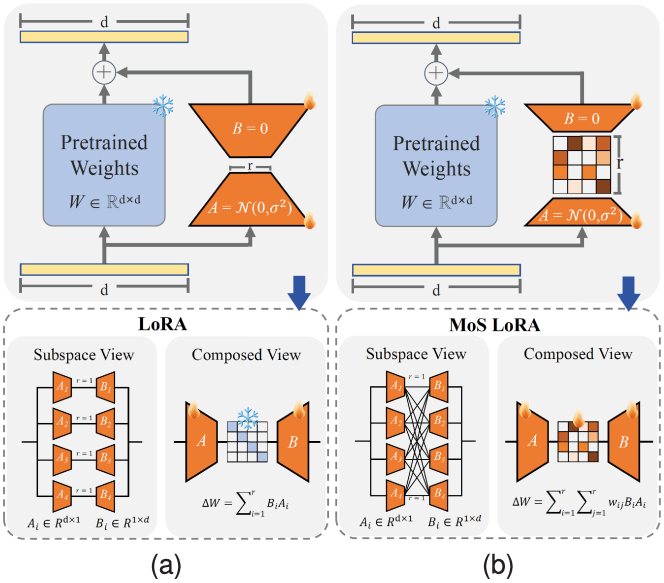



Fig. 1. Illustrations of the structures of (a) LoRA and (b) MoSLoRA.

This article proposes a novel framework based on PEFT for cross-domain HSIC. The framework incorporates the PEFT technique and the visual foundation models, which can be adapted to the cross-domain HSIC task by training only a small component using low-rank approximation. DINOv2, developed by Oquab et al. [20], is one of the newest ViT-based purely visual foundation models. It has been applied to various tasks [26], such as depth estimation in medical images [27] and change detection in remote sensing images [28], achieving excellent performance. Therefore, we apply this framework to DINOv2 and construct SpectralDINO. It consists of a source-domain preprocessing module based on source-domain spectral alignment (SDSA), a dual MoSLoRA (Dual-MoS LoRA) module with cross-domain attention (CDA), and a multilayer feature fusion (MFF) module. First, to improve

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the consistency of the spectral features extracted by the model in both the source and target domains, this article proposes an SDSA strategy, which filters the source-domain spectral information based on prior knowledge of the spectral response range, ensuring that the spectral range and number of bands in the source domains are consistent with those in the target domain. The use of aligned source-domain data limits the model's attention to bands that do not exist in the target domain, thereby encouraging the network to learn cross-domain universal features. Second, since LoRA and MoSLoRA are structured in such a way that it is difficult to distinguish single-domain specialized features from multidomain inputs, this article proposes the Dual-MoS LoRA module with CDA, which employs independent mixing matrices to learn the subspace mixer weights of the source and target domains, respectively. The CDA mechanism aims to capture the associated features of the pretrained weights, the source-domain subspaces, and the target-domain subspaces. In addition, an alternating training method is used to facilitate the learning of domain-specific knowledge through the Dual-MoS LoRA module, as well as the learning of cross-domain generic knowledge through the original LoRA matrices. Finally, MFF is employed for prototype-based FSL, maximizing the use of both shallow and deep features extracted by the model to complete the classification.

The main contributions of this article are as follows.

1) We innovatively combine the PEFT method with FSL and introduce the visual foundation models to propose a novel framework for the cross-domain HSIC task based on the PEFT method.
2) We proposed an SDSA strategy to discard information with low relevance to the spectral features of the target domain, thus promoting consistency in the extraction of spectral features by the network.
3) A novel Dual-MoS LoRA module is proposed to alleviate the difficulty of the vanilla LoRA structure to distinguish single-domain specialized features from multidomain inputs, and CDA is devised to capture the associated features of cross-domain subspace outputs.

## II. Related Works

### A. Cross-Domain FSL

Cross-domain FSL focuses on training a model using a fully labeled source-domain dataset and performing classification on a target-domain dataset with a significant shortage of labeled samples. The two primary challenges in cross-domain HSIC are the insufficient training samples in the target domain and the domain shift between different domains. To address the scarcity of training samples, researchers have explored various methods such as data augmentation, transfer learning, active learning, and weakly supervised learning [29], [30]. Among these, FSL has shown notable progress. Meta-learning, particularly episode-based training, is one of the most widely adopted strategies for FSL. In episode training, a meta-task $\mathcal{T}$ is constructed, consisting of a support set $\mathcal{S}$ and a query set $\mathcal{Q}$. The support set $\mathcal{S}$ follows the "$N$-way $K$-shot" paradigm, where it contains $N$ categories, each with $K$ samples, and the query set $\mathcal{Q}$ is used to evaluate the model's performance in the episode.

To address domain shift, current research often explores domain adaptation [11], [31], [32] and domain generalization [33], aiming to closely align the source and target domains to obtain generalized category features. Li et al. [31] were the first to address FSL and domain adaptation in a unified framework by adopting a conditional adversarial domain adaptation strategy to achieve domain distribution alignment. Zhang et al. [11] proposed a graph information aggregation cross-domain FSL (Gia-CFSL) framework to alleviate the domain shift in cross-scene classification through graph neural network information propagation and graph alignment strategies.

### B. Multidomain Learning

Existing cross-domain HSIC methods typically use a single source-domain dataset [6], [11], [13], [14], [34]. While the total number of samples in such datasets is usually sufficient compared to the limited labeled samples available in the target domain, the number of categories in a single source-domain dataset is often restricted. This limitation may result in the source domain not providing enough generalized knowledge, making it challenging for the model to extract category features with adequate discriminative properties.

Some studies have used multiple source-domain datasets [12], [15], [35] for cross-domain HSIC, but have not adequately considered the variability between the domains of different datasets. Lee et al. [36] proposed a method to train multiple hyperspectral datasets simultaneously by designing independent data-specific layers for each dataset and extracting generic features in a shared network. This method does not impose additional optimization constraints on the data-specific layers, so the features fed into the shared network may lack correlation. Yang et al. [9] designed a spectral reprojection method, which constructs a common spectral space by analyzing the wavelengths of each band of commonly used sensors, thereby achieving spectral alignment between different domains. To explore the multidomain universal representation of HSIs, Wang et al. [37] proposed multidomain networks equipped with a convolutional adaptation module to adapt to the distributional differences in different datasets, and for the first time, solved the domain shift in HSI multidomain learning.

### C. Parameter-Efficient Fine-Tuning

State-of-the-art (SOTA) deep learning foundation models often possess extremely large parameter sizes [38]. While these models perform exceptionally well across a variety of tasks, fine-tuning such large models for downstream tasks presents significant challenges for most researchers [27]. To address this issue, PEFT has been proposed first proposed in the field of natural language processing. PEFT aims to accomplish model tuning at a much lower computational cost by either training part of the existing network or by adding small-sized trainable components. PEFT methods can be broadly categorized into three types [39]: adapter-based

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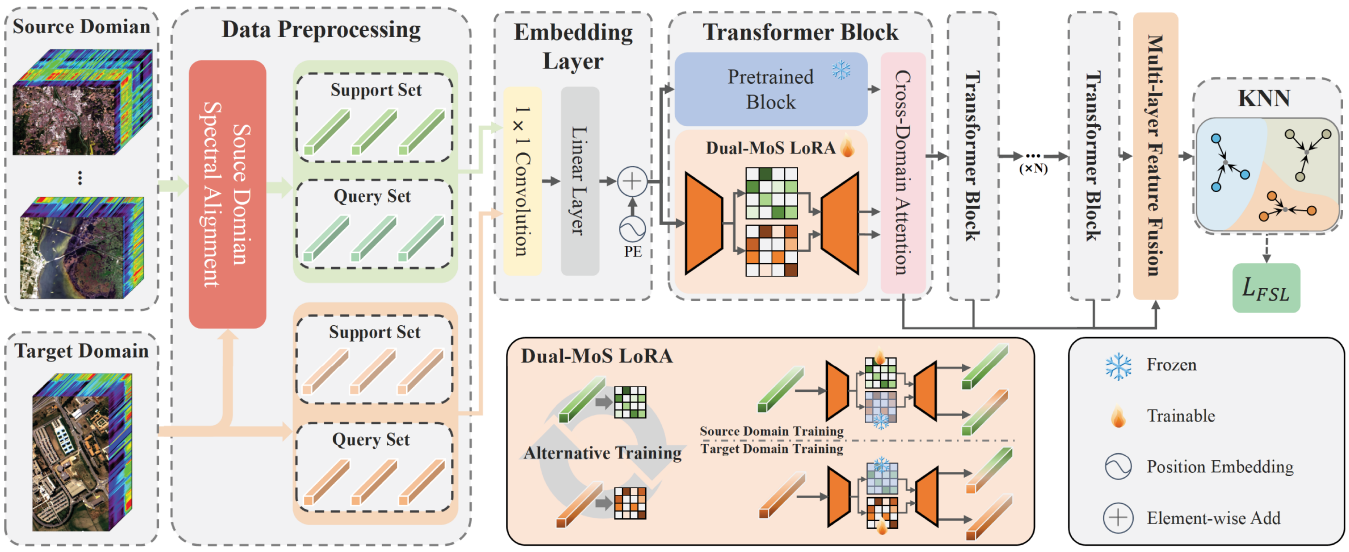



Fig. 2. Flowchart of the proposed SpectralDINO, including data preprocessing based on SDSA, Dual-MoS LoRA module with CDA, and an MFF module.

methods, which adjust the frozen network by introducing additional trainable components; prompt-based methods, which achieve fine-tuning by adding trainable tokens to the input; and LoRA [24], which approximates the update of the weight matrix through low-rank decomposition. Particularly, LoRA significantly reduces the number of parameters that need to be trained and can be seamlessly integrated with the original parameter matrices without adding any inference burden.

PEFT has also proved effective in the field of computer vision [40]. Jia et al. [41] proposed visual–prompt tuning (VPT), which applies prompt tuning to the Vision Transformer (ViT) model to adapt to downstream vision tasks. Chen et al. [42] proposed AdaptFormer, which enables models to be adapted to multiple downstream tasks in the image and video fields by fine-tuning lightweight modules. In the field of remote sensing, Xue et al. [43] proposed a segmentation method using LoRA to fine-tune SAM to accomplish the classification of airborne remote sensing images. Zhao et al. [28] compared the effects of combinations of the main PEFT methods with different foundation models, using an adapter to fine-tune DINOv2 and use it for remote sensing change detection tasks. Song et al. [44] proposed a multistage fine-tuning method to gradually guide SAM to adapt to the multispectral remote sensing crop segmentation task. PEFT is beginning to gain popularity in the domains of natural and general optical remote sensing image tasks. However, exploration of PEFT techniques in the HSIC task remains limited.

## III. Methodology

The flowchart of the proposed SpectralDINO is depicted in Fig. 2. The method encompasses three primary components: an SDSA-based data preprocessing module, Transformer blocks with a Dual-MoS LoRA module and CDA, and an MFF module. Initially, multiple source-domain datasets are aligned to match the spectral range and the number of bands of the target domain using SDSA. This step effectively minimizes spectral information in the source domain that is irrelevant to the target domain. Subsequently, the pretrained foundation model is refined with a low-rank update through the Dual-MoS LoRA module. In this stage, samples are first converted into a sequence of image tokens via the embedding layer and then processed through the Transformer block of the pretrained model and the Dual-MoS module for feature extraction. Finally, during the FSL stage, the multilayer features extracted by the model are fused through MFF, and classification is performed using a prototype-based FSL. During the training process, the network is trained alternatively by switching the source- and target-domain data as inputs.

### A. Data Preprocessing Based on SDSA

Neural networks typically require input data to have consistent dimensions, leading existing methods to unify the number of bands when processing the source and target domains. These approaches often neglect the differences in spectral information between various images, resulting in some bands from the source-domain dataset being absent in the target domain. Consequently, features extracted from these bands fail to provide cross-domain universal knowledge, thereby limiting the network's cross-domain classification capabilities.

To address this issue, we propose a spectral alignment method designed to harmonize spectral data from source domains, collected by different sensors, with the spectral range and band number of the target domain. This method filters the source-domain spectral information based on prior knowledge of the spectral response range, aiming to enhance the consistency of spectral feature extraction across both domains. This consistency is crucial for the network to effectively learn cross-domain universal knowledge from the source domain.

Given the scarcity of labeled samples in the target domain, it is essential to maintain the integrity of the target-domain data to motivate the network to make full use of the spectral information of the limited samples from the target domain. For this purpose, we only align the source-domain data with the

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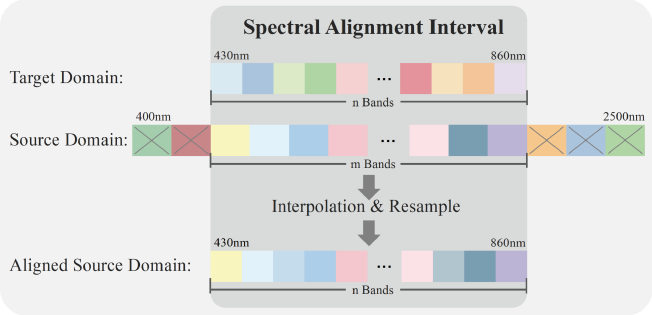



Fig. 3. Schematic of the SDSA method when the Pavia University dataset was selected for the target domain.

number of bands and the spectral range of the target-domain spectra. To this end, it is imperative that the spectral range of the source-domain dataset comprehensively encompasses that of the target domain. Typically, datasets with a spectral range of 400–2500 nm are sufficient to meet this requirement. Specifically, the spectral range of the target domain is used as the spectral alignment interval, as shown in Fig. 3. Bands of source-domain datasets outside the intervals are discarded to filter the information, and only data within the spectral alignment interval is retained to ensure the consistency of the spectral information. In addition, the source-domain data within the interval needs to be fit by interpolation and resampled to match the number of bands. We use quadratic interpolation to fit the spectral curves of the source domain and sample the data at the same location as the target-domain bands. This process ensures that the resampled source-domain data has the same spectral range and number of bands as the target domain.

Subsequently, the FSL task, denoted as an episode, is constructed following the "$N$-way, $K$-shot" paradigm. The method of constructing the FSL task is consistent for both the source and target domains. For instance, considering the source-domain dataset, $N$ categories are randomly selected from all the categories $C_s$ across multiple source-domain datasets, where $N$ corresponds to the number of categories $C_t$ present in the target domain. It is essential to ensure that $C_s > N$ to maintain the model's ability to differentiate between samples from different classes. Next, $K$ samples from each of the selected $N$ categories are chosen to form the support set $\mathcal{S}_s = \{(x_i, y_i)_{i=1}^{N\times K}\}$ and $T$ samples from each category are chosen to form the query set $\mathcal{Q}_s = \{(x_i, y_i)_{i=1}^{N\times T}\}$. Together, $\mathcal{S}_s$ and query set $\mathcal{Q}_s$ constitute an FSL task $\mathcal{T}$. In this article, following common practices in the literature [11], [12], [15], [31], we set $K$ to 1 and $T$ to 19.

### B. Feature Extraction Based on Dual-MoS LoRA

*1) Embedding Layer:* Due to the different data dimensions, HSI patches cannot directly utilize the embedding layer in foundation models designed for natural images. Therefore, it is necessary to design hyperspectral-specific embedding layers to tokenize HSI patches effectively. Some Transformer-based HSIC methods [45] tokenize HSI patches along the spectral dimension. However, given the redundancy among hyperspectral bands and the computational burden caused by the large number of tokens [46], [47], it is essential to apply dimensionality reduction to the samples before tokenization.

To extract the initial spectral features and get the appropriate number of tokens, the spectral dimensions are first downscaled using $1 \times 1$ convolution layers. Suppose that the input sample $X \in \mathbb{R}^{p\times p\times \mathrm{ch}}$, after four layers of convolution, gives the output $X' \in \mathbb{R}^{p\times p\times n}$, where $p$ represents the patch size, ch represents the spectral bands after aligned, and $n$ denotes the number of channels required for subsequent ViT Blocks (e.g., $n = 384$ for small-scale DINOv2). To transform the samples into tokens of appropriate size, the spatial dimensions also need to be linearly mapped [48]. The input sample $X$ can be tokenized by

$$E = W(g(X)) \tag{1}$$

where $E \in \mathbb{R}^{d\times n}$ collects the embedding features of the input, $W \in \mathbb{R}^{d\times p^2}$ is a trainable weight matrix and, $d$ is the dimension needed for the Transformer block, and $g(\cdot)$ represents the convolutional dimensionality reduction. Subsequently, similar to most ViT structures, a learnable class embedding $E_{\mathrm{cls}} \in \mathbb{R}^{d\times 1}$ is added to the transformed token. $E_{\mathrm{cls}}$ will learn the class features of the sample during the computation in the Transformer block and will be used for classification. The embedding token $z \in \mathbb{R}^{d\times(n+1)}$ transformed from the input samples is obtained by concatenating $E$ with $E_{\mathrm{cls}}$ and then adding it with the positional embedding $E_{\mathrm{pos}} \in \mathbb{R}^{d\times(n+1)}$, which can be calculated as

$$z = \mathrm{concat}(E_{\mathrm{cls}}, E) + E_{\mathrm{pos}}. \tag{2}$$

*2) Transformer Block:* After SDSA processing, the sample is converted into a series of image tokens in the embedding layer, which are then passed to the feature extraction network. The backbone network responsible for feature extraction is built upon the ViT architecture, which is currently one of the most popular and powerful architectures in the field of computer vision. Additionally, it integrates well with PEFT methods. Consequently, we employ a ViT-structured encoder for feature extraction. In each layer of the Transformer block, the embedding token $z \in \mathbb{R}^{d\times(n+1)}$ of the sample transformation generates query ($Q \in \mathbb{R}^{d\times(n+1)}$), key ($K \in \mathbb{R}^{d\times(n+1)}$), value ($V \in \mathbb{R}^{d\times(n+1)}$) through three projection matrices

$$Q = W_q z, K = W_k z, V = W_v z \tag{3}$$

where $W_q, W_k, W_v \in \mathbb{R}^{d\times d}$ are the learnable matrices. Subsequently, the attention output is computed by the following equation:

$$\mathrm{Attention}(Q, K, V) = \mathrm{Softmax}\left(\frac{\mathrm{QK}^T}{\sqrt{d}}\right)V. \tag{4}$$

*3) Dual-MoS LoRA:* LoRA [24] is one of the most popular PEFT methods. LoRA and its variants use a low-rank decomposition approximation to represent the update of the weight matrix. This approach greatly reduces the size of the training parameters, and the low-rank decomposition matrix can be fused with the pretrained weight matrix without increasing the inference burden of the model. To adapt foundation models for HSI few-shot classification, our methodology integrates the LoRA structure. However, vanilla LoRA architectures

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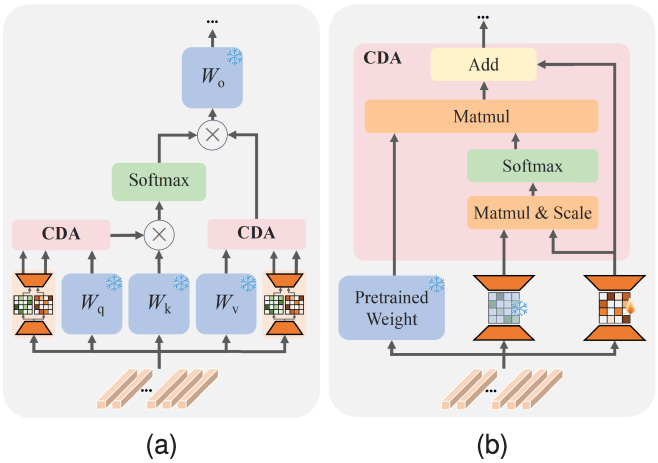



Fig. 4. (a) LoRA is only applied to $q$ and $v$ projection layers in the Transformer. (b) CDA mechanism.

have inherent structural limitations in distinguishing domain-specific features from multidomain inputs. For this reason, we propose the Dual-MoS LoRA module, which uses two independent mixing matrices to fuse the LoRA subspaces of each domain separately. Notably, these two mixing matrices use the same pair of low-rank projection matrices to ensure that they can still learn cross-domain generalized features by alternating training methods.

LoRA assumes that the update of weights has a low "intrinsic rank" during the adaptation process. Therefore, the foundation model can be adapted to the downstream task by updating the pretrained weight matrix $W_0 \in \mathbb{R}^{n\times n}$ with a low-rank decomposition. MoSLoRA sets a mixing matrix to learn the mixer weights of the fusion subspace. For the input $x$, the output $h$ obtained according to MoSLoRA can be calculated by the following equation:

$$h = W_0 x + \Delta \mathrm{W} x = W_0 x + \mathrm{B} W_m \mathrm{A} x \tag{5}$$

where $A \in \mathbb{R}^{r\times n}$, using Gaussian initialization, $B \in \mathbb{R}^{n\times r}$, using zero initialization, $W_m \in \mathbb{R}^{r\times r}$, and $r \ll n$. During the training process, the pretrained weights of the foundation model are frozen and not updated. In Dual-MoS LoRA, the mixing matrices $W_m^{\mathcal{S}} \in \mathbb{R}^{r_1\times r_1}$ and $W_m^{\mathcal{T}\in\mathbb{R}^{r_2\times r_2}}$ are set up for the source and target domains, respectively. Since they share the same pair of low-rank projection matrices, and $r_1 = r_2$, we will not distinguish between $r_1$ and $r_2$ in the following. As shown in Fig. 4(a), we follow [24], [27] to apply LoRA only to $W_q$ and $W_v$ projection layers in the Transformer block to control the size of trainable parameters.

To motivate $W_m^{\mathcal{S}}$ and $W_m^{\mathcal{T}}$ to learn the fusion subspace mixing weights of the source and target domains, respectively, we use samples from different domains to train the model alternately. Specifically, in the framework of episode training, if the source-domain samples are currently used for training, the next episode will switch to the target-domain samples for training. Accordingly, each time the source-domain samples are used for training, $W_m^{\mathcal{T}}$ is frozen so that it is not updated in the current episode and vice versa. It is worth noting that the low-rank projection matrix is involved in the whole training process for the purpose of learning cross-domain knowledge.

Although one mixing matrix is frozen in each episode, this does not affect the propagation path of the input data. As shown in Fig. 4(b), for each input sample, distinguishing between the original LoRA and the MoSLoRA, there are three outputs of the Dual-MoS LoRA, including the output of the pretrained weights and two outputs that go through $W_m^{\mathcal{S}}$ and $W_m^{\mathcal{T}}$, respectively. As can be seen from (5), the vanilla LoRA and MoSLoRA directly add the outputs of the low-rank branches to the outputs of the pretrained weights, which makes it difficult to capture the complex associations between multiple outputs. We devise a CDA mechanism to replace this operation

$$h_{\mathrm{CDA}} = \mathrm{Softmax}\left(\frac{h_s h_d^{\top}}{\sqrt{d}}\right) h_p + h_s \tag{6}$$

where $h_p$ corresponds to the output of the pretrained weights, $h_s$ is the output via the unfrozen mixing matrix, where the samples and the mixing matrix belong to the same domain, and $h_d$ is the output via the frozen mixing matrix, where the samples and the mixing matrix belong to different domains. In particular, we design the symmetry computation for alternating training and remove the projection matrix used in classical attention mechanisms, which imposes a parameter burden that exceeds that of LoRA itself.

### C. FSL Based on MFF

ViT-based networks usually contain multiple Transformer blocks. As an example, we use the "small"-scale DINOv2 in our experiments, which contains 12 Transformer blocks. The original DINOv2 only used the output of the last block. However, deep features are often highly abstract, and using only the output of the last block for classification may be limited. Some methods [27], [49] fuse the features output from different layers of the network, thus making full use of the feature information output from both shallow and deep components. Inspired by this, the proposed method designs an MFF module to fuse features extracted from Transformer blocks of different depths and uses the fused features for classification. In this article, we use the class embedding $E_{\mathrm{cls}}$ in the output of each level as the classification features. As shown in Fig. 5, in the MFF module, every three Transformer blocks are divided into a feature group $\mathcal{G}$. The outputs of all Transformer blocks within the same feature group are average-pooled to obtain the average features $f_i$

$$f_i = \frac{1}{|\mathcal{G}|}\sum_{j\in\mathcal{G}} f_i^j \tag{7}$$

where $f_i^j$ denotes the output of the $j$th Transformer block in the $i$th feature group and $|\mathcal{G}|$ denotes the number of Transformer blocks contained within the $i$th feature group. Subsequently, the average features of all the feature groups are accumulated together through learnable weights to form the fused features $f_{\mathrm{fusion}}$ that are ultimately used to perform subsequent small sample learning tasks

$$f_{\mathrm{fusion}} = \sum_i \oplus(\alpha_i f_i) \tag{8}$$

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TABLE I
DETAILS OF THE FOUR SOURCE-DOMAIN DATASETS

| Dataset | Sensor | Wavelength(nm) | Band Num. | Data Size | GSD(m) | Class Num. | Sample Num. |
|---|---|---|---|---|---|---|---|
| Augsburg | HySpex | 400-2500 | 180 | 332×485 | 30 | 7 | 78,294 |
| Berlin | HyMap | 400-2500 | 224 | 1723×476 | 13.89 | 8 | 469,671 |
| Botswana | Hyperion | 400-2500 | 145 | 1476×256 | 30 | 14 | 3248 |
| KSC | AVIRIS | 400-2500 | 176 | 512×614 | 18 | 13 | 5211 |

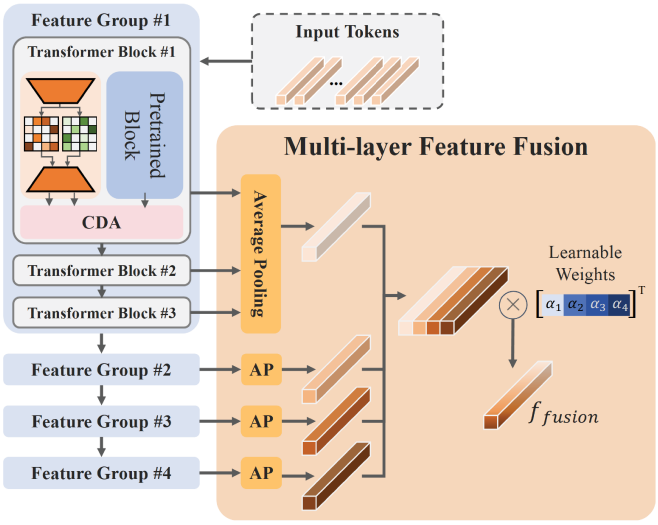



Fig. 5. Structure of the MFF module.

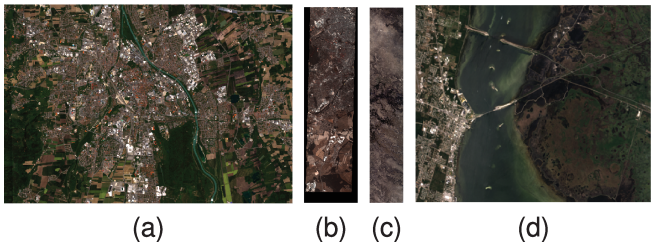



Fig. 6. Pseudo-color composite images of four source-domain datasets. (a) Augsburg. (b) Berlin. (c) Botswana. (d) KSC.

We follow [11], [13], [15] and use Euclidean distance-based prototype networks for FSL. Taking the source domain as an example, in each episode, the prototype $p_c^s$ is obtained by calculating the mean value of the features of each category in the support set. Subsequently, the fusion features $f_{\text{fusion}}^q$ of each sample $x^q$ in the query set are extracted through the network, and the Euclidean distance between $f_{\text{fusion}}^q$ and $p_c^s$ is calculated. Finally, the prediction probability $\hat{y}^q$ is predicted for each sample in the query set by the Softmax function

$$P\left(\hat{y}^q|x^q\right) = \text{Softmax}\left(-d\left(f_{\text{fusion}}^q, p_c^s\right)\right) \tag{9}$$

where $d(\cdot)$ denotes the computation of the Euclidean distance. The cross-entropy loss is used as the loss function for the network to perform FSL, which is the most common loss function in HSIC

$$L_{\text{FSL}} = -y^q \log P\left(\hat{y}^q|x^q\right). \tag{10}$$

## IV. EXPERIMENTAL RESULTS

### A. Datasets

*1) Source-Domain Datasets:* To meet the requirement of spectral alignment, this article uses datasets with a spectral range of 400–2500 nm as source-domain datasets and datasets in this spectral range as the target-domain dataset. Considering the spectral range limitations, four datasets were used as the source-domain datasets in the experiment, including Augsburg, Berlin, Botswana, and KSC. The pseudo-color composite images are shown in Fig. 6. Detailed information about these datasets is listed in Table I.

*a) Augsburg:* The Augsburg dataset [50] was acquired by the HySpex sensor over the city of Augsburg, Germany, with a spatial resolution of 30 m. The image contains 332 × 485 pixels and 180 high-quality bands.

*b) Berlin:* The Berlin dataset was processed from a simulated EnMAP dataset synthesized based on HyMap hyperspectral data with a spatial resolution of 30 m. We use the dataset processed by Hong et al. [50]. The images contain 1723 × 476 pixels, encompass 224 usable bands, with a spatial resolution of 13.89 m.

*c) Botswana:* The Botswana dataset was acquired by NASA's EO-1 satellite over the Okavango Delta, Botswana, on May 31, 2001. The raw data acquired by the Hyperion sensor consisted of 242 bands. Uncalibrated and noisy bands that cover water absorption features were removed, leaving 145 usable bands. It has 1476 × 256 pixels with a spatial resolution of 30 m.

*d) Kennedy Space Center:* The Kennedy Space Center (KSC) dataset was acquired by the NASA AVIRIS instrument on March 23, 1996, over the Kennedy Space Center, Florida. The raw data consisted of 224 bands. After removing the high noise and water absorption bands, 176 bands remain available. The dataset has 512 × 614 pixels with a spatial resolution of 18 m.

*2) Target-Domain Datasets:* The performance of each method is evaluated on target-domain datasets. In this article, the target domain includes the University of Pavia University (UP), Salinas (SA), and Houston (HT), which are commonly used datasets in hyperspectral classification tasks. The pseudo-color composite images are shown in Fig. 7.

*a) Pavia university:* The Pavia University dataset was acquired by the ROSIS sensor over Pavia, northern Italy, with a spatial resolution of 1.3 m. The image contains 610 × 340 pixels with 103 bands in the range of 430–860 nm. The dataset contains nine classes. Table II lists the details of the land-cover classes in UP.

*b) Salinas:* The Salinas dataset was acquired by the AVIRIS sensor over Salinas Valley, California, with a spatial

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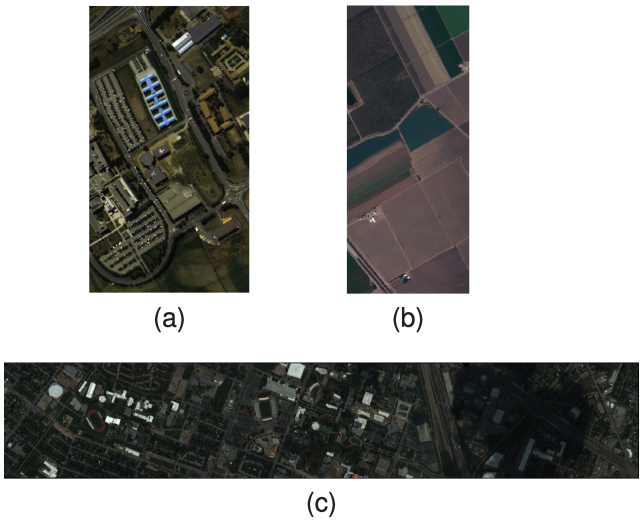



Fig. 7. Pseudo-color composite images of three target-domain datasets. (a) UP. (b) SA. (c) HT.

TABLE II
LAND-COVER CLASSES AND THE NUMBER OF SAMPLES IN UP

| Class No. | Class Name | Sample Num. |
|---|---|---|
| 1 | Asphalt | 6631 |
| 2 | Meadows | 18,649 |
| 3 | Gravel | 2099 |
| 4 | Trees | 3064 |
| 5 | Painted Metal Sheets | 1345 |
| 6 | Bare Soil | 5029 |
| 7 | Bitumen | 1330 |
| 8 | Bricks | 3682 |
| 9 | Shadows | 947 |
| | Total | 42,776 |

TABLE III
LAND-COVER CLASSES AND THE NUMBER OF SAMPLES IN SA

| Class No. | Class Name | Sample Num. |
|---|---|---|
| 1 | Brocoli_green_weeds 1 | 2009 |
| 2 | Brocoli_green_weeds 2 | 3726 |
| 3 | Fallow | 1976 |
| 4 | Fallow_rough_plow | 1394 |
| 5 | Fallow_smooth | 2678 |
| 6 | Stubble | 3959 |
| 7 | Celery | 3579 |
| 8 | Grape_untrained | 11,271 |
| 9 | Soil_vinyard_develop | 6203 |
| 10 | Corn_senesced_green_weeds | 3278 |
| 11 | Lettuce_romaine_4wk | 1068 |
| 12 | Lettuce_romaine_5wk | 1927 |
| 13 | Lettuce_romaine_6wk | 916 |
| 14 | Lettuce_romaine_7wk | 1070 |
| 15 | Vinyard_untrained | 7268 |
| 16 | Vinyard_vertical_trellis | 1807 |
| | Total | 54,129 |

resolution of 3.7 m. As with other AVIRIS-acquired images, removing bands covering the region of water absorption leaves 204 usable bands in the 400–2500-nm range. The dataset contains 512 × 217 pixels with 16 classes. Table III lists the details of the feature classes in this dataset.

TABLE IV
LAND-COVER CLASSES AND THE NUMBER OF SAMPLES IN HT

| Class No. | Class Name | Sample Num. |
|---|---|---|
| 1 | Healthy grass | 1363 |
| 2 | Stressed grass | 1366 |
| 3 | Synthetic grass | 760 |
| 4 | Trees | 1355 |
| 5 | Soil | 1353 |
| 6 | Water | 354 |
| 7 | Residential | 1382 |
| 8 | Commercial | 1355 |
| 9 | Road | 1364 |
| 10 | Highway | 1337 |
| 11 | Railway | 1345 |
| 12 | Parking Lot 1 | 1343 |
| 13 | Parking Lot 2 | 511 |
| 14 | Tennis Court | 466 |
| 15 | Running Track | 719 |
| | Total | 16,373 |

*c) Houston:* The Houston dataset was acquired by the CASI-1500 airborne sensor in 2012 over the University of Houston campus and the surrounding urban area with a spatial resolution of 2.5 m. The dataset contains 144 bands in the 380–1050-nm range. It spans 349 × 1905 pixels and provides 15 land-cover classes. Table IV lists the details of the land-cover classes in HT.

### B. Experimental Setting

To facilitate the learning of the subspaces mixing weights of different domains by the Dual-MoS LoRA module, an alternating training method is adopted in the training of SpectralDINO. The model is trained using the source dataset and the target dataset in turn, with the training data in each episode being different from that in the previous episode. Specifically, 200 labeled samples are selected from each class of the source-domain dataset, with the number of labeled samples greater than 200, to form the training dataset. For the target-domain dataset, 5 labeled samples are randomly selected for each category, and the number is increased to 200 by data augmentation. Rectangles with an aspect ratio of 3:4–4:3 and an area range of 8%–100% of the original input are used to randomly crop the patch and then restore it to the original patch size. Thus, the source-domain training dataset contains $m \times 200$ samples, where $m$ is the total number of categories in the source-domain datasets with more than 200 labeled samples. The target-domain training dataset contains $n \times 200$ samples, where $n$ is the number of categories contained in the target-domain dataset. A total of 37 valid categories were filtered out of the four source-domain datasets selected in the experiment.

All experiments are performed on a server with an NVIDIA GeForce RTX 3090 GPU and 512 GB of memory. The input patch size is set to $9 \times 9$. SpectralDINO was implemented with PyTorch. Adam is used as the optimizer, and the number of training iterations is set to 3000. The proposed method uses an initial learning rate of $5e^{-4}$, and the weight decay factor is set to $5e^{-2}$.

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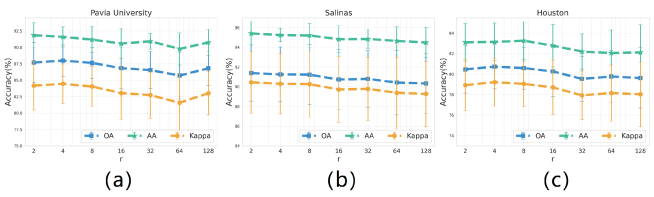



Fig. 8. Testing the effect of $r$ on the classification performance of the proposed method on three datasets. (a) Pavia University. (b) Salinas. (c) Houston.

## C. Parameter Analysis

In this section, the performance of the proposed method is tested under different parameter settings. In the application of LoRA, different $r$ affects the adaptation performance in downstream tasks and the size of trainable parameters. Additionally, it offers insights into the correlation between cross-domain data to a certain extent. Therefore, the setting of $r$ is crucial when using LoRA to adjust the model with low-rank approximation. This section will test the effect of different $r$ on SpectralDINO. In addition, the number of embedding tokens can be controlled by adjusting the spectral dimensions. The number of tokens also affects the efficiency of extracting features. Therefore, the rank $r$ of the Dual-MoS LoRA module and the number of tokens are considered as adjustable hyperparameters. To reduce the pressure of hyperparameter optimization, suboptimal parameters are used. The values of $r$ are selected from $\{2, 4, 8, 16, 32, 64, 128\}$ and the number of tokens is selected from $\{1, 2, 4, 8, 16, 32, 64\}$. Each combination was randomly tested 10 times to select the best combination.

The performance of SpectralDINO under various $r$ settings across the three datasets is illustrated in Fig. 8. The overall accuracy (OA), average category accuracy (AA), and Kappa coefficient are employed to evaluate the classification outcomes. It can be observed that as $r > 8$, the model's performance on the three datasets begins to deteriorate with the continued increase of $r$, indicating that a smaller $r$ is typically sufficient for model adaptation. An excessively large $r$ may force LoRA to amplify more features from the pretrained weights, which are often superfluous for downstream tasks. Due to the domain shift, the same source domain provides inconsistent universal knowledge to the model for different target domains. Consequently, an appropriate value of $r$ should be selected for each dataset. Based on the results presented in Fig. 8, we set $r$ to 4 for UP and HT and set $r$ to 2 for SA.

We follow the analysis of subspace similarity in [24] and use a normalized subspace similarity based on the Grassmann distance for the subspace similarity of the projection matrix $A$ of the last transformer block

$$\phi\left(A_{r=8}, A_{r=32}, i, j\right) = \frac{\left\| U_{A_{r=8}}^{i\top} U_{A_{r=32}}^{j} \right\|_F^2}{\min\left(i, j\right)} \in [0, 1] \tag{11}$$

where $A_{r=8} \in \mathbb{R}^{8\times n}$ and $A_{r=32} \in \mathbb{R}^{32\times n}$ are the low-rank projection matrices $A$ of rank $r = 8$ and $r = 32$ obtained by learning using the same pretrained model, and $U_{A_{r=8}}$ and $U_{A_{r=32}}$ are their right-singular matrices after singular value decomposition. $U_{A_{r=8}}^{i}$ denotes the column corresponding to the top-$i$ singular values of $U_{A_{r=8}}$. The result of (11) is 1 when the subspaces are completely overlapped and 0 when they are completely separated.

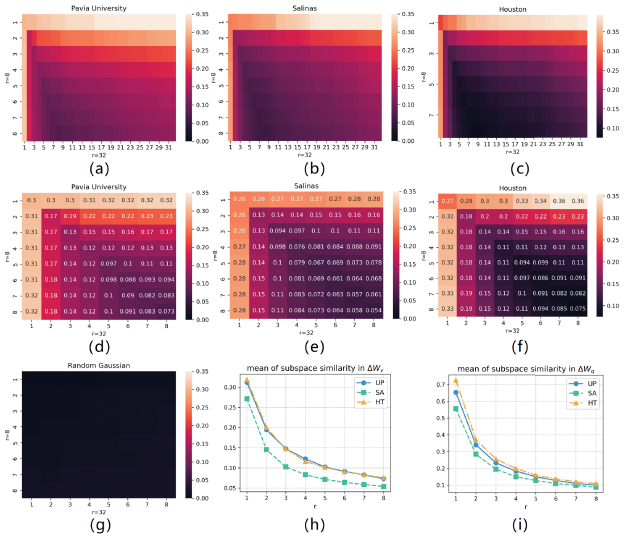



Fig. 9. Subspace similarity between singular vectors of $A_{r=8}$ and $A_{r=32}$ for $\Delta W_v$ on (a) UP, (b) SA, and (c) HT. Localization with values on (d) UP, (e) SA, and (f) HT. (g) Same heatmap between two random Gaussian matrices. Mean value of similarity for the $i$th row and column on (h) $\Delta W_v$. A similar trend can be seen in (i) $\Delta W_q$ as well.

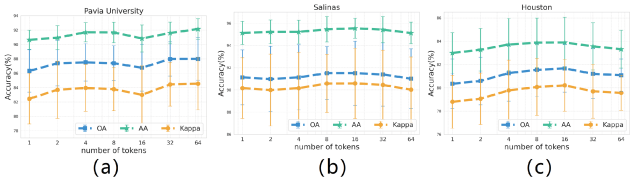



Fig. 10. Testing the effect of the number of tokens on the classification performance on three datasets. (a) Pavia University. (b) Salinas. (c) Houston.

Heatmaps of subspace similarity in $\Delta W_v$ are demonstrated in Fig. 9(a)–(c). It can be seen that $\Delta W_v$ on UP and HT appears to have a higher "intrinsic rank" than SA. Specifically, in the local images with numerical values, we find that the mean value of the $i$th row and column in UP is approximately equal to that in HT and higher than that in SA. This trend is consistent with the $r$ we chose for these datasets. As a comparison, Fig. 9(g) shows the subspace similarity between two random Gaussian matrices of the same size as $A_{r=8}$ and $A_{r=32}$. The same situation also occurs in $\Delta W_q$, as shown in Fig. 9(i).

Fig. 10 demonstrates the performance of SpectralDINO under different token number configurations. It is evident that as the number of tokens increases, the model's performance on each dataset exhibits an overall upward trend, indicating that an increase in token number results in a richer feature representation, which, in turn, enhances the model's performance. Nevertheless, the computational and memory burden also escalates rapidly with the increase in token numbers. We note that the model's capability improves swiftly when the number of tokens increases from 1 to 8. When reaching 8 tokens, the classification accuracy improvement rate on the three datasets began to slow down or even decline. Based on the results presented in Fig. 10, to balance the model's

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TABLE V
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE UP DATASET

| Class | SVM | 3D CNN | SSRN | Gia-CFSL | CMFSL | RPCL-FSL | FDFSL | FSCF-SSL | SpectralDINO |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 60.20 | 62.19 | 83.32 | 80.68 | 83.41 | 81.96 | 83.20 | 78.62 | **93.75** |
| 2 | 75.12 | 67.75 | 64.48 | **88.17** | 83.89 | 82.80 | 81.31 | 87.25 | 84.23 |
| 3 | 24.02 | 53.90 | 60.56 | 57.93 | 67.55 | 70.80 | 63.80 | 81.02 | **85.01** |
| 4 | 91.00 | 80.96 | 94.79 | 93.61 | 93.16 | 93.89 | 90.62 | **96.25** | 95.12 |
| 5 | 92.11 | 91.47 | 99.68 | 99.03 | 99.40 | 99.53 | 99.43 | **99.99** | 99.34 |
| 6 | 14.37 | 42.87 | **93.60** | 74.06 | 87.82 | 77.03 | 71.56 | 85.37 | 83.66 |
| 7 | 79.46 | 80.83 | 95.15 | 80.03 | 82.78 | 85.43 | 89.92 | 85.18 | **98.23** |
| 8 | 76.37 | 62.32 | 85.81 | 72.08 | 73.69 | 83.91 | 86.40 | **96.32** | 90.20 |
| 9 | **99.77** | 92.68 | 98.64 | 97.59 | 99.50 | 98.94 | 93.26 | 99.42 | 96.26 |
| OA(%) | 65.62±4.57 | 65.46±5.64 | 77.45±4.84 | 83.17±2.55 | 84.06±3.43 | 83.25±2.13 | 81.80±2.12 | 87.41±3.94 | **88.15±2.03** |
| AA(%) | 68.05±3.35 | 70.55±2.56 | 86.23±1.89 | 82.58±1.64 | 85.69±1.78 | 86.03±2.20 | 84.39±1.38 | 89.93±3.14 | **91.76±1.38** |
| Kappa | 55.57±4.41 | 56.49±6.08 | 72.07±5.39 | 78.09±3.07 | 79.50±4.20 | 78.41±2.47 | 76.49±2.29 | 83.68±4.88 | **84.67±2.53** |

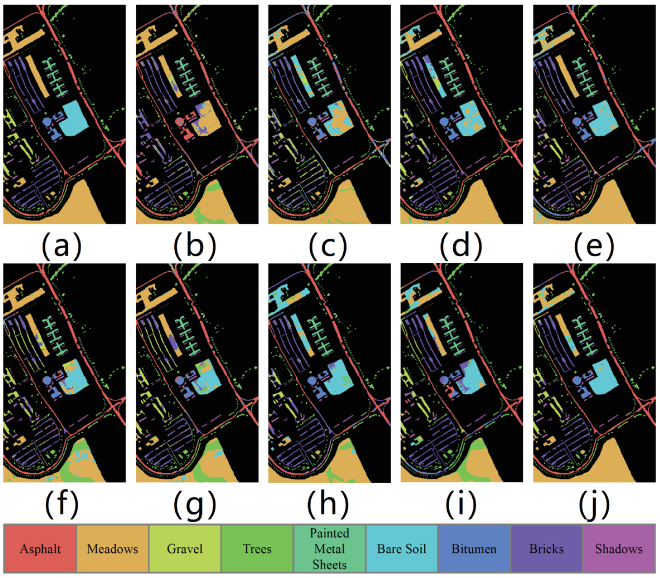



Fig. 11. Classification maps of different methods on the UP dataset, including (a) ground truth, (b) SVM, (c) 3-D CNN, (d) SSRN, (e) Gia-CFSL, (f) CMFSL, (g) RPCL-FSL, (h) FDFSL, (i) FSCF-SSL, and (j) SpectralDINO.

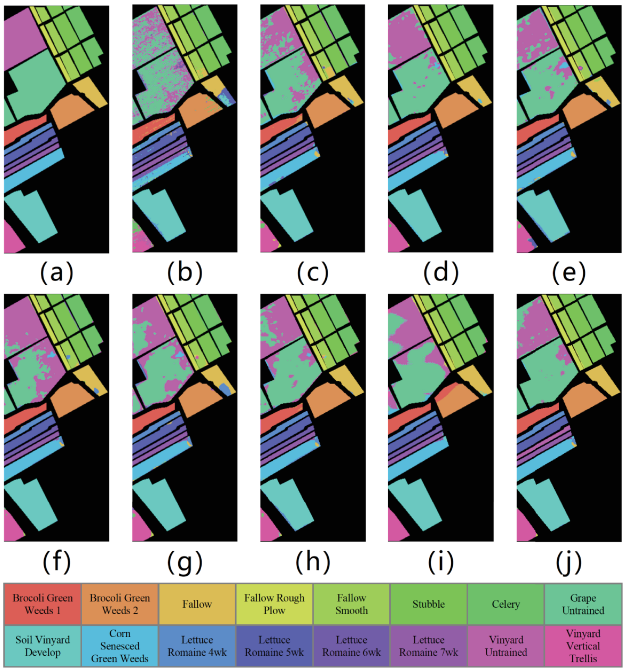



Fig. 12. Classification maps of different methods on the SA dataset, including (a) ground truth, (b) SVM, (c) 3-D CNN, (d) SSRN, (e) Gia-CFSL, (f) CMFSL, (g) RPCL-FSL, (h) FDFSL, (i) FSCF-SSL, and (j) SpectralDINO.

performance with computational load, we set the number of tokens to 8 for UP and SA, and 16 for HT.

### D. Compared With Other Methods

To evaluate the performance of SpectralDINO in cross-domain few-shot classification, some SOTA deep learning and FSL-based methods are used for comparison, including the SVM, 3DCNN [51], SSRN [52], Gia-CFSL [11], CMFSL [12], RPCL-FSL [13], FDFSL [6], and FSCF-SSL [53]. SVM, 3DCNN, and SSRN are machine learning- and deep learning-based methods. Since such methods require the input data to have the same dimensionality, five labeled samples of each class from the target domain are selected as the training set, and the rest of the samples are used for testing. Gia-CFSL, CMFSL, RPCL-FSL, and FDFSL are FSL-based methods. FSCF-SSL is based on self-supervised learning (SSL). For these FSL-based and SSL-based methods, we follow the settings in the original literature to conduct the experiments, including the selection of the source-domain dataset, the data preprocessing, the construction of meta-learning tasks, and the settings of various parameters to ensure the fairness of the experiments. All experiments were conducted with five randomly selected labeled samples from each category of the target domain for training. OA, AA, and Kappa coefficients are used to evaluate the classification effectiveness of different methods. For easy observation and comparison, the kappa coefficients in this article are all 100 times the original values. We use common random partitioning for all datasets, following all the FSL-based methods. To ensure the reliability of the results, all methods are run independently ten times on each dataset, and the average results are reported.

Table V lists the classification results of the different methods on the UP dataset. SpectralDINO achieved the best result with an OA of 88.15%. Overall, the classification performance of FSL-based methods is generally better than deep learning-based methods. Fig. 11 visualizes the performance of different methods on the UP dataset. The OA of FSL- and SSL-based methods can generally reach more than 80%, which is at least 4.35% better than deep learning-based methods, and the performance is more stable. This reflects the advantage of FSL

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TABLE VI
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE SA DATASET

| Class | SVM | 3DCNN | SSRN | Gia-CFSL | CMFSL | RPCL-FSL | FDFSL | FSCF-SSL | SpectralDINO |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 98.30 | 95.95 | 98.68 | 99.12 | 99.51 | 99.55 | 98.32 | **99.82** | 99.71 |
| 2 | 73.87 | 97.66 | **99.93** | 99.55 | 99.87 | 99.84 | 98.70 | 97.82 | 99.68 |
| 3 | 41.69 | 91.95 | 95.14 | 89.12 | **97.30** | 90.49 | 89.03 | 92.87 | 95.79 |
| 4 | 97.71 | 98.52 | 99.09 | 99.48 | 99.55 | 99.18 | 99.34 | **99.70** | 99.27 |
| 5 | 95.40 | 93.61 | 91.92 | 89.32 | 92.78 | 92.45 | 92.36 | **97.27** | 93.77 |
| 6 | 98.30 | 97.41 | 99.26 | 98.21 | 98.59 | 97.54 | **99.33** | 99.28 | 98.36 |
| 7 | 99.40 | 99.17 | **99.85** | 96.28 | 99.22 | 98.79 | 99.25 | 98.78 | 99.55 |
| 8 | 64.23 | 66.85 | 70.43 | 73.86 | 75.06 | 80.53 | **81.54** | 79.81 | 79.21 |
| 9 | 97.45 | 97.27 | 98.57 | 97.52 | 97.42 | **99.83** | 98.38 | 99.57 | 99.67 |
| 10 | 27.74 | 77.82 | 83.34 | 83.71 | 86.95 | 86.49 | 84.41 | **94.92** | 91.35 |
| 11 | 84.26 | 89.39 | 97.78 | 96.85 | 99.14 | 98.78 | 96.30 | 99.42 | **99.69** |
| 12 | 97.14 | 99.10 | **99.79** | 99.12 | 99.51 | 99.29 | 99.66 | 98.81 | 99.07 |
| 13 | 98.98 | **99.82** | 99.59 | 98.40 | 98.95 | 98.96 | 98.05 | 99.05 | 96.67 |
| 14 | 87.13 | 94.30 | 95.75 | 97.83 | 98.75 | 97.24 | **99.52** | 96.36 | 99.12 |
| 15 | 40.70 | 61.59 | 78.58 | 75.18 | 81.05 | 77.12 | 82.04 | 76.38 | **83.43** |
| 16 | 75.21 | 88.97 | 94.39 | 87.90 | 95.93 | 90.55 | 91.93 | 95.84 | **97.59** |
| OA(%) | 73.96±2.59 | 84.35±2.03 | 88.74±2.78 | 88.00±1.90 | 90.31±1.51 | 90.56±1.49 | 91.17±1.14 | 91.29±1.47 | **92.02±2.31** |
| AA(%) | 79.84±2.67 | 90.59±1.28 | 93.88±1.42 | 92.59±1.06 | 94.97±0.68 | 94.16±0.93 | 94.26±1.31 | 95.36±1.11 | **95.75±0.90** |
| Kappa | 71.04±2.92 | 82.62±2.25 | 87.51±3.09 | 86.68±2.08 | 89.24±1.66 | 89.51±1.64 | 90.18±1.26 | 90.31±1.63 | **91.14±2.54** |

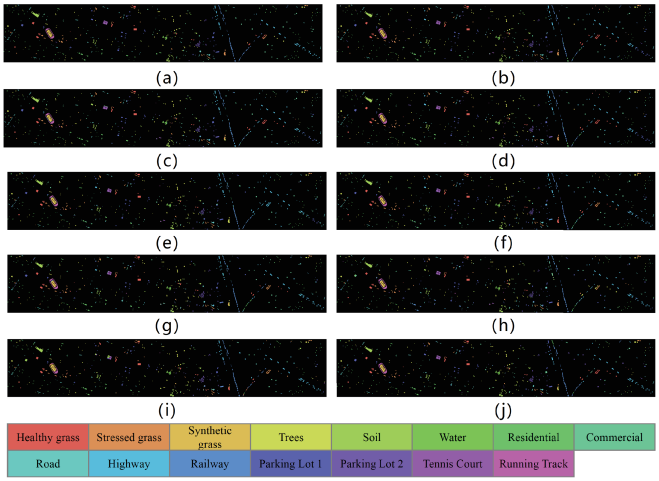



Fig. 13. Classification maps of different methods on the HT dataset, including (a) ground truth, (b) SVM, (c) 3-D CNN, (d) SSRN, (e) Gia-CFSL, (f) CMFSL, (g) RPCL-FSL, (h) FDFSL, (i) FSCF-SSL, and (j) SpectralDINO.

and SSL in the case of an insufficient sample. SpectralDINO achieved the best classification results on Classes 1, 3, and 7 (i.e., "asphalt," "gravel," and "bitumen"). Compared to other methods, the accuracy of these classes improved by at least 10.34%, 3.99%, and 3.08%, respectively.

Table VI lists the classification results of the different methods on the SA dataset. Our method continues to hold a competitive edge across all three metrics. Fig. 12 shows the classification results of the different methods. For the SA dataset, each method could accomplish classification well, with a total of seven methods achieving more than 88% OA. Our method achieves the best accuracy in three categories and the second-highest accuracy in six categories. Notably, SpectralDINO achieves the best accuracy in Class 15, "Vinyard_untrained," which has more than 7000 samples, and is also the fourth best in Class 8, "Grape_untrained," with an accuracy of nearly 80%. This shows that the proposed method can extract more representative discriminative features from a limited number of samples.

Table VII lists the classification results of different methods on the HT dataset. RPCL-FSL has achieved the second-best classification results. It is noteworthy that both SSRN and RPCL-FSL demonstrate optimal performance in four categories. SSRN demonstrates superior performance in comparison to several FSL-based methods and exhibits notable generalization capabilities. Overall, SpectralDINO achieves the best classification performance. It outperforms the suboptimal method by 2.03%, 1.71%, and 2.18 for OA, AA, and kappa, respectively. Fig. 13 shows the visualization of the classification results of each method on the HT dataset.

### E. Ablation Experiment

The proposed framework consists of data preprocessing based on SDSA, feature extraction based on Dual-MoS LoRA, and an MFF module. To verify the contribution and validity of the individual components, we performed an ablation analysis by removing individual components from the entire framework. There are five variants of the ablation analysis as follows.

1) *SpectralDINO (Without SDSA)*: source domains are not processed using SDSA. To keep the spectral dimensions the same, we use PCA to unify different datasets to the same number of bands.
2) *SpectralDINO (With MoSLoRA)*: replace the Dual-MoS LoRA with a MoSLoRA.
3) *SpectralDINO (With LoRA)*: replace the Dual-MoS LoRA with a LoRA.
4) *SpectralDINO (Without CDA)*: remove the CDA. Simply add the output of the current domain branch to the pretrained model.
5) *SpectralDINO (Without MFF)*: remove the MFF and use only the output of the last Transformer block for FSL classification.

The mean and standard deviation of the ten experiments for the different ablation models are presented in Table VIII. It can be found that SpectralDINO outperforms the other ablation

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TABLE VII
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE HT DATASET

| Class | SVM | 3DCNN | SSRN | Gia-CFSL | CMFSL | RPCL-FSL | FDFSL | FSCF-SSL | SpectralDINO |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 77.33 | 88.99 | 89.22 | 88.66 | 89.85 | **89.86** | 87.24 | 80.39 | 89.28 |
| 2 | 48.14 | 81.91 | 78.96 | 81.91 | 86.64 | **86.98** | 87.85 | 54.66 | 86.13 |
| 3 | 98.48 | 90.75 | **98.57** | 94.87 | 98.37 | 94.91 | 97.75 | 97.46 | 96.40 |
| 4 | 86.29 | 89.68 | 92.96 | 91.40 | 90.72 | **93.12** | 91.38 | 88.03 | 89.52 |
| 5 | 80.27 | 96.80 | 94.96 | 94.96 | 94.32 | 97.78 | **99.32** | 89.87 | 98.13 |
| 6 | 83.31 | 77.69 | **89.94** | 75.53 | 84.28 | 86.75 | 85.06 | 89.84 | 88.63 |
| 7 | 22.70 | 68.61 | 71.68 | 62.97 | 73.31 | 79.50 | 69.44 | **81.01** | 77.47 |
| 8 | 17.12 | 40.40 | 36.62 | 43.72 | 44.20 | 40.35 | **49.90** | 46.09 | 47.69 |
| 9 | 73.98 | 68.49 | 75.47 | 63.81 | 72.18 | 71.59 | 72.17 | 71.68 | **77.66** |
| 10 | 8.27 | 66.64 | 67.95 | 66.10 | 61.81 | 68.89 | 64.12 | 79.07 | **84.27** |
| 11 | 43.53 | 64.97 | 70.39 | 65.27 | 68.70 | 69.02 | 67.36 | **81.93** | 75.13 |
| 12 | 14.06 | 51.56 | 54.21 | **70.48** | 63.38 | 69.34 | 62.17 | 67.09 | 65.67 |
| 13 | 10.09 | 87.74 | 90.99 | 89.87 | 93.04 | **96.59** | 94.25 | 88.62 | 86.49 |
| 14 | 94.23 | 93.00 | **100.00** | 92.77 | 91.94 | 91.75 | 97.28 | 94.07 | 99.72 |
| 15 | 94.64 | 93.45 | **99.80** | 96.06 | 97.31 | 96.20 | 97.21 | 89.92 | 96.21 |
| OA(%) | 52.63±1.91 | 74.88±2.65 | 77.27±3.05 | 76.10±1.14 | 77.92±1.54 | 79.64±1.98 | 78.57±1.87 | 77.12±3.50 | **81.67±2.05** |
| AA(%) | 56.83±1.44 | 77.38±2.22 | 80.78±2.48 | 78.56±1.28 | 80.67±1.41 | 82.18±1.84 | 81.50±1.48 | 79.98±3.62 | **83.89±2.17** |
| Kappa | 49.16±2.03 | 72.90±2.84 | 75.49±3.28 | 74.20±1.22 | 76.15±1.67 | 78.03±2.13 | 76.86±2.01 | 75.29±3.80 | **80.21±2.22** |

TABLE VIII
PERFORMANCE OF DIFFERENT ABLATION MODELS ON EACH DATASET

| Dataset | | w/o SDSA | w/ MoSLoRA | w/ LoRA | w/o CDA | w/o MFF | SpectralDINO |
|---|---|---|---|---|---|---|---|
| Pavia University | OA(%) | 72.81 ± 5.24 | 87.46 ± 2.83 | 87.53 ± 2.87 | 87.01 ± 3.19 | 87.55 ± 2.53 | **88.15 ± 2.03** |
| | AA(%) | 77.10 ± 3.34 | 91.31 ± 1.95 | 91.05 ± 1.83 | 90.48 ± 2.31 | **91.98 ± 1.64** | 91.76 ± 1.38 |
| | Kappa | 65.92 ± 6.05 | 83.85 ± 3.54 | 83.90 ± 3.64 | 83.22 ± 4.02 | 83.99 ± 3.16 | **84.67 ± 2.53** |
| Salinas | OA(%) | 90.07 ± 1.02 | 91.21 ± 2.52 | 90.55 ± 2.50 | 91.16 ± 2.59 | 91.34 ± 2.95 | **92.02 ± 2.31** |
| | AA(%) | 94.68 ± 0.52 | 95.08 ± 1.12 | 94.63 ± 0.98 | 94.96 ± 1.31 | 95.50 ± 1.22 | **95.75 ± 0.90** |
| | Kappa | 88.98 ± 1.12 | 90.25 ± 2.77 | 89.51 ± 2.74 | 90.19 ± 2.86 | 90.39 ± 3.25 | **91.14 ± 2.54** |
| Houston | OA(%) | 75.77 ± 3.96 | 80.30 ± 2.29 | 80.80 ± 2.18 | 81.09 ± 1.44 | 81.40 ± 1.91 | **81.67 ± 2.05** |
| | AA(%) | 78.97 ± 3.51 | 82.83 ± 2.25 | 83.36 ± 1.99 | 83.47 ± 1.47 | 83.57 ± 1.65 | **83.89 ± 2.17** |
| | Kappa | 73.83 ± 4.29 | 78.73 ± 2.47 | 79.28 ± 2.34 | 79.58 ± 1.55 | 79.92 ± 2.05 | **80.21 ± 2.22** |

variants on three datasets. Specifically, after removing SDSA, the proposed method shows a significant decrease in OA, AA, and Kappa, with the most substantial impact observed on the UP dataset. We note that the spectral ranges of the SA datasets are consistent with all source-domain datasets, indicating that spectral range alignment was applied on UP and HT. This could account for the significant decline in the network's performance on UP and HT following the removal of SDSA. After the removal of SDSA, the information outside the spectral range of the target domain was input into the network following PCA processing, resulting in a decrease in network performance. Replacing Dual-MoS LoRA with MoSLoRA and LoRA resulted in a decrease in the model's OA by at least 0.62%, 0.81%, and 0.87% across all three datasets. Similarly, the removal of CDA also led to a decline in performance, indicating that integrating the outputs from all branches can yield a more effective feature representation. After the removal of MFF, the network experienced a slight increase in AA on the UP dataset due to the improved precision in some classes with fewer samples. However, the removal also caused a significant decrease in both OA and Kappa metrics.

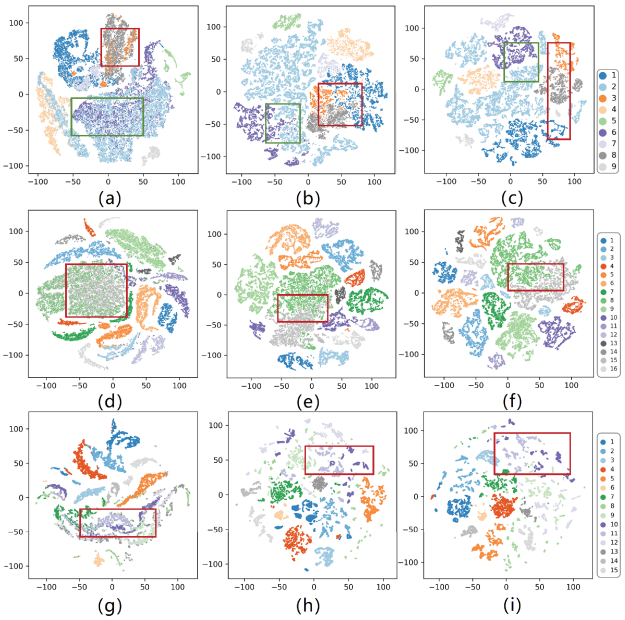



Fig. 14. Visualization of feature separability. (a) Original samples, (b) RPCL-FSL, and (c) SpectralDINO for UP. (d) Original samples, (e) RPCL-FSL, and (f) SpectralDINO for SA. (g) Original samples, (h) RPCL-FSL, and (i) SpectralDINO for HT.

### F. Feature Visualization

To intuitively reflect the effectiveness of the features extracted by different methods, we use the t-SNE method to visualize the original samples and the features extracted by RPCL-FSL and SpectralDINO on the three datasets. Fig. 14

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TABLE IX
NUMBER OF TRAINABLE PARAMETERS, FLOPS, AND COMPUTATION TIME FOR DIFFERENT METHODS ON THE THREE DATASETS

| | Methods | Gia-CFSL | CMFSL | RPCL-FSL | FDFSL | FSCF-SSL | SpectralDINO |
|---|---|---|---|---|---|---|---|
| UP | Trainable Params | 0.2087M | 0.6140M | **0.0720M** | 0.1834M | 3.1562M | 0.2597M |
| | FLOPs | 384.5668M | **244.1883M** | 766.6675M | 469.2563M | 1797.8418M | 1869.3938M |
| | Training Time | 3466.35s | 579.28s | 661.37s | **344.20s** | 821.64s | 540.83s |
| | Testing Time | 10.86s | **2.66s** | 18.69s | 9.85s | 52.02s | 19.91s |
| SA | Trainable Params | 0.2188M | 0.6140M | **0.0827M** | 0.1834M | 3.3106M | 0.2427M |
| | FLOPs | 683.6744M | **434.1124M** | 1362.9824M | 834.2334M | 3210.7235M | 3338.8462M |
| | Training Time | 4747.14s | 917.29s | 897.49s | **534.81s** | 1361.02s | 760.31s |
| | Testing Time | 10.41s | **2.99s** | 16.51s | 15.72s | 38.34s | 25.29s |
| HT | Trainable Params | 0.2128M | 0.6140M | **0.0766M** | 0.1834M | 3.2182M | 0.2717M |
| | FLOPs | 640.9447M | **406.9804M** | 1277.7936M | 782.0938M | 3001.8956M | 5461.8509M |
| | Training Time | 3280.79s | 881.46s | **362.78s** | 431.82s | 1064.72s | 662.78s |
| | Testing Time | 1.22s | **0.89s** | 4.40s | 6.38s | 6.47s | 9.53s |

illustrates the 2-D projected distribution of these samples and features. The original samples of the UP and HT are mixed severely. Observing Fig. 14(a), the distributions of Class 3 (orange) and Class 8 (dark gray) in the red box and Class 2 (light blue) and Class 6 (dark purple) in the green box are nearly completely overlapped. This is because samples from different categories may have similar spectra, while there may also be large differences in the spectra of samples of the same class [54]. The SA has the least mixing of the original samples. As can be seen in Fig. 14(d), many categories are well clustered and separable, but near-complete overlap also occurs in Class 8 (green) and Class 15 (gray) in the red box. Separation of these samples by SpectralDINO is significantly better than the other two methods. Our method exhibited distinct clustering effects and clear interclass distances in the feature projection across all three datasets. For example, in the UP, the features extracted by RPCL-FSL for Class 1 (dark blue), Class 3 (orange), and Class 8 (dark gray) show some mixing. In contrast, the features extracted by SpectralDINO for Class 3 (orange) and Class 8 (dark gray) show much less mixing, and both remain sufficiently separable from the features of Class 1 (dark blue).

### G. Parameters and Computational Complexity

In this section, we analyze the computational complexity of the comparison methods and the proposed SpectralDINO. Table IX presents the training time, testing time, number of trainable parameters, and floating-point operations (FLOPs) for Gia-CFSL, CMFSL, RPCL-FSL, FDFSL, FSCF-SSL, and SpectralDINO across the three target-domain datasets. Overall, SpectralDINO exhibits a moderate number of trainable parameters and training time compared to other FSL- and SSL-based methods. We employed the small-scale DINOv2. For instance, the model's total parameter count in the HT dataset reaches 21.3688 M.

Consequently, the FLOPs of SpectralDINO are significantly higher than those of the other methods, resulting in a longer testing time on each dataset. During the training phase, the number of trainable parameters of our method can be minimized to 0.2427 M on the UP dataset, which constitutes only 1.14% of the total model parameters. This accounts for why the training time does not increase substantially compared to other methods. Notably, our method has the second shortest training time on UP and SA. The results indicate that Dual-MoS LoRA can efficiently perform low-rank approximation tuning of the foundation model, significantly reducing the fine-tuning cost.

## V. CONCLUSION

In this article, we developed SpectralDINO, utilizing the PEFT method to adapt the visual foundation model DINOv2 for the cross-domain few-shot HSIC task. First, to ensure the extraction of consistent spectral features across different datasets, an SDSA strategy is employed to preprocess the source domain data. This approach enables the model to focus more on the spectral information relevant to the target domain, thereby facilitating the extraction of more generalized features. Second, to alleviate the issue of the LoRA structure's difficulty in extracting domain-specific features from multidomain inputs, we have designed a dual mixture-of-subspace LoRA module in the feature extraction phase. This method promotes the network to learn domain-specific knowledge by alternately training two independent mixture matrices, utilizing different subspace mixing weights, while maintaining a low computational cost. Finally, multilayer fused features are employed to accomplish few-shot classification, fully leveraging the shallow and deep features extracted by the network. Experimental results on three datasets confirm the effectiveness of the proposed method. While SpectralDINO has demonstrated promising results, there is room for further research. For instance, exploring the performance of different combinations of PEFT methods and various foundation models in cross-domain few-shot HSIC could be a promising direction for future research.

## REFERENCES

[1] A. Rizaldy, A. J. Afifi, P. Ghamisi, and R. Gloaguen, "Improving mineral classification using multimodal hyperspectral point cloud data and multi-stream neural network," *Remote Sens.*, vol. 16, no. 13, p. 2336, Jun. 2024.

[2] G. Avola, A. Matese, and E. Riggi, "An overview of the special issue on 'precision agriculture using hyperspectral images,"' *Remote Sens.*, vol. 15, no. 7, p. 1917, Jan. 2023.

[3] J. G. A. Barbedo, "A review on the combination of deep learning techniques with proximal hyperspectral images in agriculture," *Comput. Electron. Agricult.*, vol. 210, Jul. 2023, Art. no. 107920.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:53:27 UTC from IEEE Xplore. Restrictions apply.

[4] K. Tan, L. Chen, H. Wang, Z. Liu, J. Ding, and X. Wang, "Estimation of the distribution patterns of heavy metal in soil from airborne hyperspectral imagery based on spectral absorption characteristics," *J. Environ. Manage.*, vol. 347, Dec. 2023, Art. no. 119196.

[5] X. Wang, J. Liu, W. Chi, W. Wang, and Y. Ni, "Advances in hyperspectral image classification methods with small samples: A review," *Remote Sens.*, vol. 15, no. 15, p. 3795, Jul. 2023.

[6] B. Qin, S. Feng, C. Zhao, W. Li, R. Tao, and W. Xiang, "Cross-domain few-shot learning based on feature disentanglement for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5514215.

[7] Y. Peng, Y. Liu, B. Tu, and Y. Zhang, "Convolutional transformer-based few-shot learning for cross-domain hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1335–1349, 2023.

[8] X. Lu, T. Gong, and X. Zheng, "Domain mapping network for remote sensing cross-domain few-shot classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5606411.

[9] B. Yang, S. Hu, Q. Guo, and D. Hong, "Multisource domain transfer learning based on spectral projections for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3730–3739, 2022.

[10] S. Jia, S. Jiang, Z. Lin, N. Li, M. Xu, and S. Yu, "A survey: Deep learning for hyperspectral image classification with few labeled samples," *Neurocomputing*, vol. 448, pp. 179–204, Aug. 2021.

[11] Y. Zhang, W. Li, M. Zhang, S. Wang, R. Tao, and Q. Du, "Graph information aggregation cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 2, pp. 1–14, Feb. 2024.

[12] B. Xi, J. Li, Y. Li, R. Song, D. Hong, and J. Chanussot, "Few-shot learning with class-covariance metric for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 5079–5092, 2022.

[13] Q. Liu et al., "Refined prototypical contrastive learning for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5506214.

[14] J. Li, Z. Zhang, R. Song, Y. Li, and Q. Du, "SCFormer: Spectral coordinate transformer for cross-domain few-shot hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 33, pp. 840–855, 2024.

[15] B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan, and R. Wang, "Deep few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2290–2304, Apr. 2019.

[16] Z. Xue, Y. Zhou, and P. Du, "S3Net: Spectral–spatial Siamese network for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5531219.

[17] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16000–16009.

[18] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 8748–8763.

[19] A. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 4015–4026.

[20] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," 2023, *arXiv:2304.07193*.

[21] B. X. Yu et al., "Visual tuning," *ACM Comput. Surv.*, vol. 56, no. 12, pp. 1–38, Jul. 2024.

[22] L. Scheibenreif, M. Mommert, and D. Borth, "Parameter efficient self-supervised geospatial domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27841–27851.

[23] N. Houlsby et al., "Parameter-efficient transfer learning for NLP," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 2790–2799.

[24] E. J. Hu et al., "LoRA: Low-rank adaptation of large language models," 2021, *arXiv:2106.09685*.

[25] T. Wu, J. Wang, Z. Zhao, and N. Wong, "Mixture-of-subspaces in low-rank adaptation," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, Nov. 2024, pp. 7880–7899.

[26] F. Chen, M. V. Giuffrida, and S. A. Tsaftaris, "Adapting vision foundation models for plant phenotyping," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2023, pp. 604–613.

[27] B. Cui, M. Islam, L. Bai, and H. Ren, "Surgical-DINO: Adapter learning of foundation models for depth estimation in endoscopic surgery," *Int. J. Comput. Assist. Radiol. Surgery*, vol. 19, no. 6, pp. 1013–1020, Mar. 2024.

[28] Y. Zhao, Y. Zhang, Y. Dong, and B. Du, "Adapting vision transformer for efficient change detection," 2023, *arXiv:2312.04869*.

[29] N. Wambugu et al., "Hyperspectral image classification on insufficient-sample and feature learning using deep neural networks: A review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 105, Dec. 2021, Art. no. 102603.

[30] M. A. Moharram and D. M. Sundaram, "Land use and land cover classification with hyperspectral data: A comprehensive review of methods, challenges and future directions," *Neurocomputing*, vol. 536, pp. 90–113, Jun. 2023.

[31] Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li, and Q. Du, "Deep cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501618.

[32] X. Zheng, H. Cui, and X. Lu, "Multiple source domain adaptation for multiple object tracking in satellite video," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5626911.

[33] Y. Zhang, W. Li, W. Sun, R. Tao, and Q. Du, "Single-source domain expansion network for cross-scene hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 32, pp. 1498–1512, 2023.

[34] H. Wang, X. Wang, and Y. Cheng, "Graph meta transfer network for heterogeneous few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501112.

[35] H. Lee, S. Eum, and H. Kwon, "Cross-domain CNN for hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 3627–3630.

[36] H. Lee, S. Eum, and H. Kwon, "Is pretraining necessary for hyperspectral image classification?," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 3321–3324.

[37] J. Wang, J. Zhou, and X. Liu, "Toward universal representation learning for multidomain hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508116.

[38] S. Hayou, N. Ghosh, and B. Yu, "LoRA+: Efficient low rank adaptation of large models," 2024, *arXiv:2402.12354*.

[39] S.-Y. Liu et al., "DoRA: Weight-decomposed low-rank adaptation," 2024, *arXiv:2402.09353*.

[40] Z. Zhong, Z. Tang, T. He, H. Fang, and C. Yuan, "Convolution meets LoRA: Parameter efficient finetuning for segment anything model," 2024, *arXiv:2401.17868*.

[41] M. Jia et al., "Visual prompt tuning," in *Proc. Eur. Conf. Comput. Vis.*, Oct. 2022, pp. 709–727.

[42] S. Chen et al., "AdaptFormer: Adapting vision transformers for scalable visual recognition," in *Proc. NIPS*, 2022, pp. 16664–16678.

[43] B. Xue, H. Cheng, Q. Yang, Y. Wang, and X. He, "Adapting segment anything model to aerial land cover classification with low-rank adaptation," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[44] B. Song, H. Yang, Y. Wu, P. Zhang, B. Wang, and G. Han, "A multispectral remote sensing crop segmentation method based on segment anything model using multistage adaptation fine-tuning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4408818.

[45] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

[46] X. Zheng, Y. Yuan, and X. Lu, "Dimensionality reduction by spatial–spectral preservation in selected bands," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 5185–5197, Sep. 2017.

[47] X. Zheng, H. Sun, X. Lu, and W. Xie, "Rotation-invariant attention network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 4251–4265, 2022.

[48] Y. Wang et al., "Spectral–spatial–temporal transformers for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536814.

[49] J. Feng et al., "CNN-based multilayer spatial–spectral feature fusion and sample augmentation with local and nonlocal constraints for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 4, pp. 1299–1313, Apr. 2019.

[50] D. Hong, J. Hu, J. Yao, J. Chanussot, and X. X. Zhu, "Multimodal remote sensing benchmark datasets for land cover classification with a shared and specific feature learning model," *ISPRS J. Photogramm. Remote Sens.*, vol. 178, pp. 68–80, Aug. 2021.

[51] Y. Li, H. Zhang, and Q. Shen, "Spectral–spatial classification of hyperspectral imagery with 3D convolutional neural network," *Remote Sens.*, vol. 9, no. 1, p. 67, Jan. 2017.

[52] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[53] Z. Li et al., "Few-shot hyperspectral image classification with self-supervised learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514215.

[54] J. Yao, D. Hong, C. Li, and J. Chanussot, "SpectralMamba: Efficient mamba for hyperspectral image classification," 2024, *arXiv:2404.08489*.


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